R3.2 Electron transfer reactions

Electroplating (HL)

Higher level (HL)

Learning outcomes

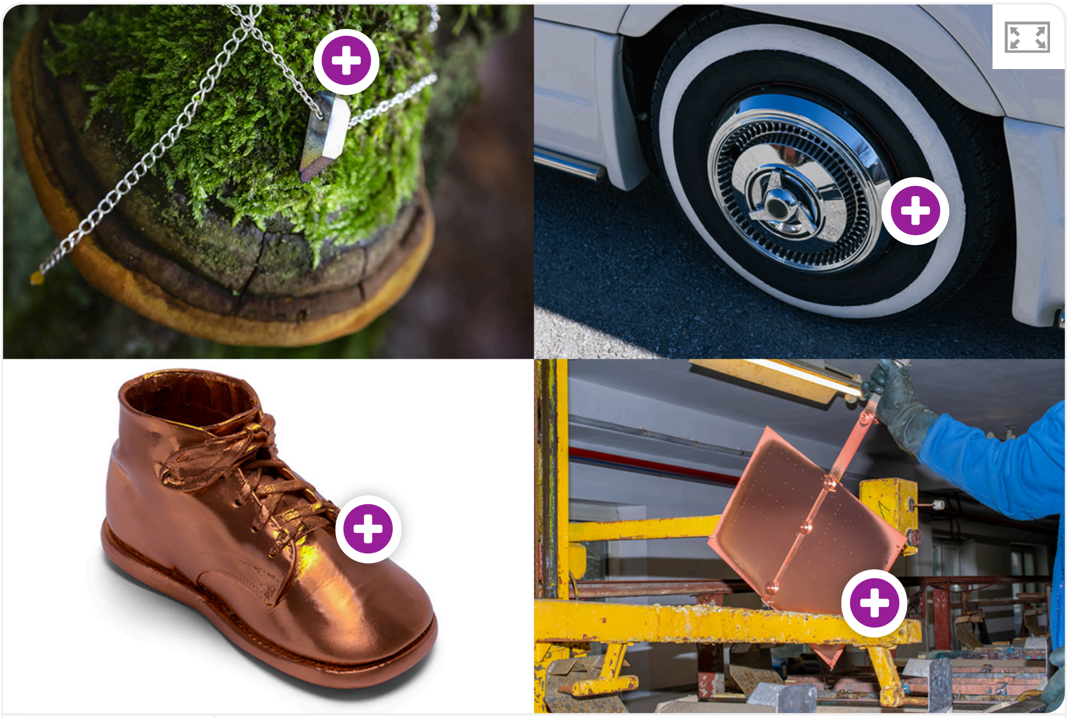
By the end of this section you should be able to deduce equations for the electrode reactions during electroplating.

Electroplating: Depositing metallic ions on surfaces

Many of the objects we have around us are made of materials that have been chemically coated or plated in a thin coat of solid metal. Sometimes this is done to improve the appearance of the object, or to improve its physical properties like resilience, lustre, or hardness.

Take for example gold tone jewellery. Have you ever purchased a somewhat inexpensive pair of earrings just to notice that after a few months the gold seems to be wearing off and another metal is being exposed underneath or has left a greenish stain on your skin? This is an example of electroplating one metal onto another. But how does this occur? Is this thin coating of metal simply painted on, or is it somehow bonded to the material underneath?

Use **Interactive 1** to investigate some of the materials that are used in electroplating.



Rights of use

Interactive 1. Electroplating Different Objects.

More information for interactive 1

A collage of four images, each featuring an object with a plus icon indicating a hotspot which when clicked, pops up additional information.

The top left image is a silver chain necklace with a pendant resting on a moss-covered tree trunk. The hotspot reads, Crystal quartz electroplated on a silver chain.

The top right image is a close-up of a car tire with a white sidewall and a shiny chrome hubcap. The hotspot reads, A chrome hubcap.

The bottom left image is a baby shoe coated in a reflective copper finish. The hotspot reads, A bronzed baby shoe.

The bottom right image is a worker handling a copper-plated industrial component in a workshop setting, with machinery and metal structures in the background. The hotspot reads: A copper electroplating machine for printed circuit boards.

The answer lies in how electroplating occurs. Electroplating involves the use of an electrolysis reaction, driving a non-spontaneous electrochemical reaction through the use of a power supply. Once connected to the power supply, the process allows for one metal to be transferred within a solution onto the surface of an electrode.

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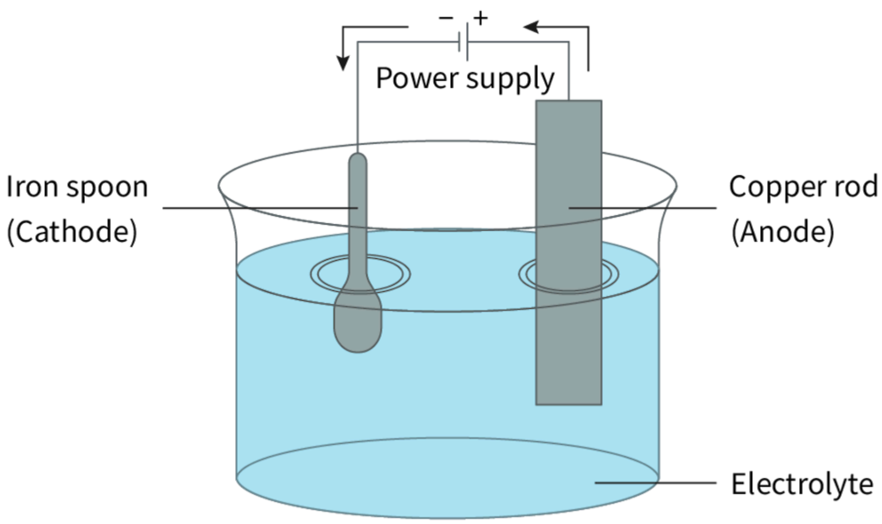


Figure 1. Plating an iron spoon with copper.

More information for figure 1

The image is a diagram of an electroplating setup. It shows a cylindrical container filled with a blue solution labeled as 'Electrolyte.' Inside the container, there is an iron spoon on one side labeled as 'Iron spoon (Cathode).' On the opposite side, there is a copper rod labeled as 'Copper rod (Anode).' Both the spoon and the rod are partially submerged in the solution but extend above it.

Above the container, there's a representation of a power supply. The spoon is connected to the negative terminal and the copper rod to the positive terminal, suggesting the flow of current. Arrows indicate the direction of current from the power supply to the electrodes. This setup is used to plate the iron spoon with a layer of copper transferred from the copper rod through the electrolysis process.

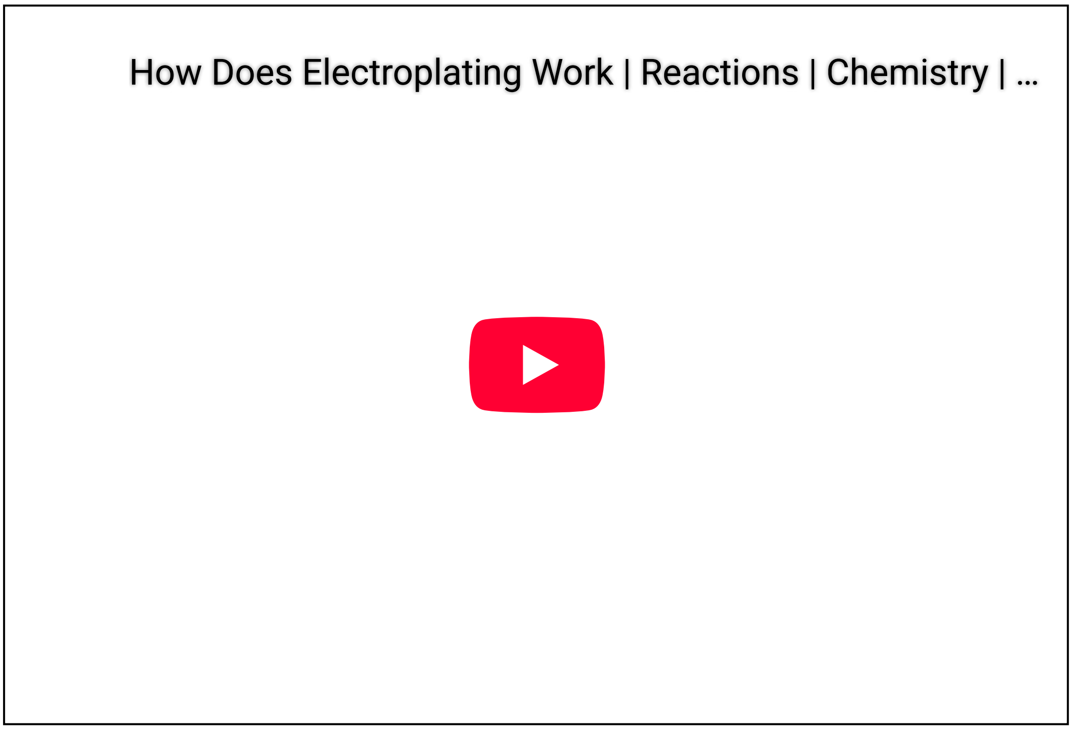
[Generated by AI]

An electroplating system must consist of:

- a power supply (as the reaction is non-spontaneous)
- a conductive cathode (the object you wish to plate)
- a metal anode
- an electrolytic solution containing a common metal ion to the anode.

The cathode can be any solid object so long as it is electrically conductive. This electrode can be made of an inert material like graphite or platinum or a reactive metal.

When the reaction is assembled and the current is connected, electrons flow from the power supply into the cathode, just as in any other electrolysis reaction. The metal cations in solution then migrate to the cathode where they collect the electrons on the surface of the electrode and bond to the cathode as a thin layer of metal atoms. This process of layering a thin sheet of atoms of one metal, through reduction, over a surface is known as plating. If the cathode is made of metal then the plated material is alloyed onto the surface of the metal electrode.



Video 1. Electroplating reactions in action.

As the plating process occurs and metallic ions are reduced and pulled out of solution to be deposited on the cathode, the anode slowly starts to undergo an oxidation reaction. If the anode is made of a reactive metal then the anode is slowly eroded, as it is oxidised to form metallic ions in solution.

You may recall that when inert electrodes like platinum or graphite are used in electrolysis reactions, the oxygen in the water molecules from the aqueous solution will be preferentially reduced and you will see oxygen gas being evolved according to the reaction:

$$\mathrm{H_2O(l) \rightarrow \frac{1}{2}O_2(g) + 2H^+(g) + 2e^-} \quad E^{\ominus}_{\mathrm{ox}} = -1.23\ \mathrm{V}$$

In electroplating reactions, the anode is a reactive metal and is preferentially oxidised over the water molecules.

Worked example 1

Stainless-steel (an alloy of carbon and iron) is often plated with copper for decorative purposes. The setup involves a solution of copper(II) sulfate, a copper electrode, a stainless steel object you wish to plate, and a power supply.

1. State which electrode is the anode and which is the cathode in this reaction.
2. Give the half-reaction that is allowing for the plating of the stainless-steel electrode.
3. Describe the process that is occurring at the anode, using a half-reaction.
4. Describe two experimental techniques that would allow you to observe that a chemical reaction is occurring.

1. The stainless steel electrode acts as the cathode and a copper electrode would be needed to act as the anode.
2. $$Cu^{2+}(aq) + 2e^{-} \rightarrow Cu(s)$$
3. The anode is the location of the oxidation half-reaction, and the copper electrode becomes pitted or eroded.

$$Cu(s) \rightarrow Cu^{2+}(aq) + 2e^{-}$$

4. Change in mass of the cathode and anode.
   Change in colour of the electrolyte solution.

Activity

- **IB learner profile attribute:**
  - Inquirer
  - Reflective
- **Approaches to learning:**
  - Thinking skills — Combining different ideas in order to create new understandings
  - Research skills — Comparing, contrasting and validating information, Using search engines and libraries effectively
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Pair/group activity

Electroplating is a highly ubiquitous chemical process. In your daily life you come across hundreds of examples of electroplated items without even realising it. These items are often overlooked and how they are created is sometimes poorly understood by students and adults alike.

Your task

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In this activity you will research to identify an object that you use in your daily life that is fabricated using electroplating. You will want to determine, and then explain, how this particular object is manufactured using electroplating.

It may be interesting to choose an object that is manufactured in your region or community. You may wish to extend your research to include the environmental impact of electroplating and a description of how solutions that contain zinc and other heavy metal ions are toxic and have to be disposed of in an appropriate manner.

Sketch a simple model showing the set-up for this type of electroplating reaction and give the equations for the half reactions taking place at each electrode.

Extension

If you were interested in performing a similar electroplating experiment using the lab equipment available in your school, what methods would you take to quantitatively measure the rate or yield of the reaction?

Determine the theoretical cell potential and change in Gibbs free energy for this reaction.